

BOOTSTRAP

Bootstrap Structure and Files

Exercise 2.1:

Explore the structure of the downloaded Bootstrap directory. Identify and explain the purpose of the CSS, JS, and icons folders.

Exercise 2.2:

Modify your HTML to include Bootstrap's JavaScript plugins via bootstrap.bundle.min.js.

Source Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Bootstrap via CDN Method</title>


  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

</head>

<body class="bg-light text-dark">


  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
    <div class="container">
      <a class="navbar-brand" href="#">CDN Project</a>

      <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarNav">

        <span class="navbar-toggler-icon"></span>

      </button>

      <div class="collapse navbar-collapse" id="navbarNav">
        <ul class="navbar-nav ms-auto">
          <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
          <li class="nav-item"><a class="nav-link" href="#">Features</a></li>
        </ul>
      </div>
    </div>
  </nav>
```

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```
<div class="container my-5">

  <div class="p-5 mb-4 bg-white border rounded-3 shadow-sm">

    <h1 class="display-5 fw-bold text-success">Success! Bootstrap is Loaded via CDN</h1>

    <p class="col-md-8 fs-4 text-muted">You are successfully rendering this page using direct
links to Bootstrap's hosting servers. No local assets required.</p>

    <hr class="my-4">

    <button class="btn btn-primary btn-lg" type="button" data-bs-toggle="modal" data-bs-
target="#exampleModal">

      Launch Interactive Modal

    </button>

  </div>

</div>

<div class="modal fade" id="exampleModal" tabindex="-1" aria-
labelledby="exampleModalLabel" aria-hidden="true">

  <div class="modal-dialog">

    <div class="modal-content">

      <div class="modal-header">

        <h5 class="modal-title" id="exampleModalLabel">Interactive Plugin Check</h5>

        <button type="button" class="btn-close" data-bs-dismiss="modal" aria-
label="Close"></button>

      </div>

      <div class="modal-body">

        If you can see this popup box, it proves that the
<strong>bootstrap.bundle.min.js</strong> script file is working perfectly!

      </div>

      <div class="modal-footer">

        <button type="button" class="btn btn-secondary" data-bs-
dismiss="modal">Close</button>

      </div>

    </div>

  </div>

</div>

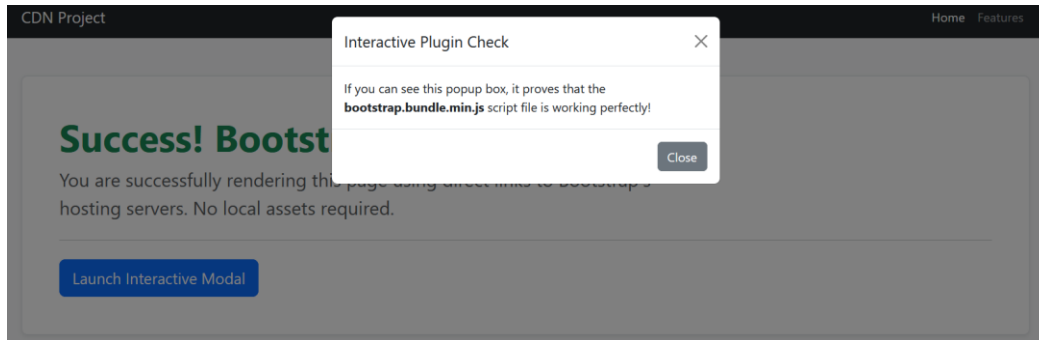
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
```

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</body>

</html>

Output:



Fundamentals of Responsive Grid Layout

Exercise 3.1:

Create a container with three columns that stack vertically on mobile, two-per-row on tablets, and three-per-row on desktops.

Exercise 3.2:

Use .container, .row, and .col-* classes appropriately for responsive design.

Source Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Fundamentals of Responsive Grid Layout</title>

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

  <style>
    .grid-box {
      background-color: #0d6efd;
      color: white;
      padding: 20px;
      text-align: center;
      border-radius: 8px;
      font-weight: bold;
      box-shadow: 0 4px 6px rgba(0,0,0,0.1);
    }
  </style>
</head>
<body class="bg-light">

  <div class="container my-5">
```

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```
<header class="text-center mb-5">
  <h1 class="display-5 fw-bold text-dark">Exercise 3.1 & 3.2: Responsive Grid</h1>
  <p class="text-muted">Resize your browser window to watch the columns shift layouts
automatically!</p>
  <hr>
</header>

<div class="row g-4">

  <div class="col-12 col-md-6 col-lg-4">
    <div class="grid-box">
      Column 1
      <div class="small fw-normal mt-1 d-none d-lg-block">Desktop: Width 4/12</div>
      <div class="small fw-normal mt-1 d-none d-md-block d-lg-none">Tablet: Width
6/12</div>
      <div class="small fw-normal mt-1 d-md-none">Mobile: Width 12/12</div>
    </div>
  </div>

  <div class="col-12 col-md-6 col-lg-4">
    <div class="grid-box bg-success">
      Column 2
      <div class="small fw-normal mt-1 d-none d-lg-block">Desktop: Width 4/12</div>
      <div class="small fw-normal mt-1 d-none d-md-block d-lg-none">Tablet: Width
6/12</div>
      <div class="small fw-normal mt-1 d-md-none">Mobile: Width 12/12</div>
    </div>
  </div>

  <div class="col-12 col-md-6 col-lg-4">
    <div class="grid-box bg-warning text-dark">
      Column 3
      <div class="small fw-normal mt-1 d-none d-lg-block">Desktop: Width 4/12</div>
      <div class="small fw-normal mt-1 d-none d-md-block d-lg-none">Tablet: Width 6/12
(Wrapped)</div>
      <div class="small fw-normal mt-1 d-md-none">Mobile: Width 12/12</div>
    </div>
  </div>

</div>

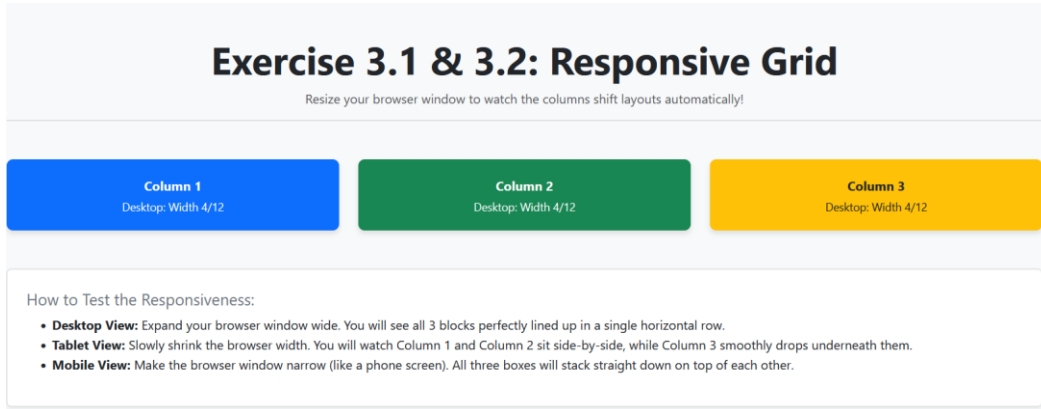
<div class="mt-5 card p-4 bg-white shadow-sm">
  <h3 class="h5 text-secondary">How to Test the Responsiveness:</h3>
  <ul>
    <li><strong>Desktop View:</strong> Expand your browser window wide. You will see all
3 blocks perfectly lined up in a single horizontal row.</li>
    <li><strong>Tablet View:</strong> Slowly shrink the browser width. You will watch
Column 1 and Column 2 sit side-by-side, while Column 3 smoothly drops underneath them.</li>
    <li><strong>Mobile View:</strong> Make the browser window narrow (like a phone
screen). All three boxes will stack straight down on top of each other.</li>
  </ul>
</div>

</div>
```

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```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Output:



Column Layouts and Grid Classes

Exercise 4.1:

Design a two-column layout with a sidebar (col-md-3) and content area (col-md-9).

Exercise 4.2:

Create a four-column layout (col-sm-3) with equal width.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Column Layouts and Grid Classes</title>

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

  <style>
    /* Custom helper styles to make the layouts clearly visible */
    .demo-box {
      padding: 20px;
      text-align: center;
      font-weight: bold;
      border-radius: 6px;
      margin-bottom: 15px;
      box-shadow: inset 0 0 0 2px rgba(0,0,0,0.05);
    }
    .sidebar-bg { background-color: #ffc107; color: #000; } /* Yellow */
    .content-bg { background-color: #0dcaf0; color: #000; } /* Cyan */
    .equal-col-bg { background-color: #198754; color: #fff; } /* Green */
  </style>
</head>
<body class="bg-light">

  <div class="container my-5">
```

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```
<header class="text-center mb-5">
  <h1 class="display-5 fw-bold">Exercise 4: Column Layouts & Grid Classes</h1>
  <p class="text-muted">Mastering asymmetric sidebars and equal-width multi-column
structures.</p>
  <hr>
</header>

<section class="mb-5">
  <h2 class="h4 mb-3 text-secondary">Exercise 4.1: Sidebar (3) & Content (9) Layout</h2>
  <p class="small text-muted">Behavior: Side-by-side on medium screens (tablets/desktops)
and stacks vertically on mobile screens.</p>

  <div class="row">
    <div class="col-md-3">
      <div class="demo-box sidebar-bg">
        Sidebar <br>
        <span class="fw-normal small">.col-md-3</span>
      </div>
    </div>

    <div class="col-md-9">
      <div class="demo-box content-bg">
        Main Content Area <br>
        <span class="fw-normal small">.col-md-9</span>
      </div>
    </div>
  </div>
</section>

<section class="mb-5">
  <h2 class="h4 mb-3 text-secondary">Exercise 4.2: Four Equal Width Columns</h2>
  <p class="small text-muted">Behavior: Aligns into 4 columns even on small devices
(landscape phones), breaking into individual rows only on extra-small mobile viewports.</p>

  <div class="row g-3">
    <div class="col-sm-3">
      <div class="demo-box equal-col-bg">
        Col 1 <br>
        <span class="fw-normal small">.col-sm-3</span>
      </div>
    </div>

    <div class="col-sm-3">
      <div class="demo-box equal-col-bg">
        Col 2 <br>
        <span class="fw-normal small">.col-sm-3</span>
      </div>
    </div>

    <div class="col-sm-3">
      <div class="demo-box equal-col-bg">
        Col 3 <br>
        <span class="fw-normal small">.col-sm-3</span>
      </div>
    </div>
  </div>
```

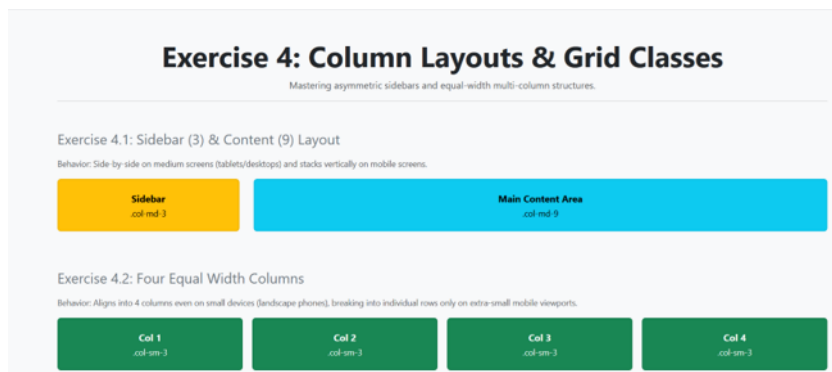
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```
</div>

<div class="col-sm-3">
  <div class="demo-box equal-col-bg">
    Col 4 <br>
    <span class="fw-normal small">.col-sm-3</span>
  </div>
</div>
</div>
</section>

</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```



Buttons

Exercise 9.1:

Create buttons using all contextual classes: btn-primary, btn-secondary, btn-outline-*, etc.

Exercise 9.2:

Add button groups using btn-group, and create toggle buttons with checkboxes.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Buttons & Button Groups</title>

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="bg-light">

  <div class="container my-5">

    <header class="text-center mb-5">
      <h1 class="display-5 fw-bold">Exercise 9: Buttons and Button Groups</h1>
      <p class="text-muted">Exploring styling variants, button clusters, and state control
mechanisms.</p>
```

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```
<hr>
</header>
```

```
<section class="mb-5 p-4 bg-white border rounded shadow-sm">
  <h2 class="h4 mb-3 text-secondary">Exercise 9.1: Contextual & Outline Buttons</h2>
```

```
  <h5 class="text-muted mt-3 mb-2">Solid Theme Variants</h5>
```

```
  <div class="d-flex flex-wrap gap-2 mb-4">
    <button type="button" class="btn btn-primary">Primary</button>
    <button type="button" class="btn btn-secondary">Secondary</button>
    <button type="button" class="btn btn-success">Success</button>
    <button type="button" class="btn btn-danger">Danger</button>
    <button type="button" class="btn btn-warning">Warning</button>
    <button type="button" class="btn btn-info">Info</button>
    <button type="button" class="btn btn-light">Light</button>
    <button type="button" class="btn btn-dark">Dark</button>
    <button type="button" class="btn btn-link">Link</button>
  </div>
```

```
  <h5 class="text-muted mb-2">Outline Variants (<code>.btn-outline-*</code>)</h5>
```

```
  <div class="d-flex flex-wrap gap-2">
    <button type="button" class="btn btn-outline-primary">Primary</button>
    <button type="button" class="btn btn-outline-secondary">Secondary</button>
    <button type="button" class="btn btn-outline-success">Success</button>
    <button type="button" class="btn btn-outline-danger">Danger</button>
    <button type="button" class="btn btn-outline-warning">Warning</button>
    <button type="button" class="btn btn-outline-info">Info</button>
    <button type="button" class="btn btn-outline-dark">Dark</button>
  </div>
</section>
```

```
<section class="mb-5 p-4 bg-white border rounded shadow-sm">
  <h2 class="h4 mb-3 text-secondary">Exercise 9.2: Button Groups & Checkbox Toggles</h2>
```

```
  <h5 class="text-muted mt-3 mb-2">Standard Button Group (<code>.btn-group</code>)</h5>
```

```
  <div class="btn-group mb-4" role="group" aria-label="Basic example">
    <button type="button" class="btn btn-primary">Left</button>
    <button type="button" class="btn btn-primary">Middle</button>
    <button type="button" class="btn btn-primary">Right</button>
  </div>
```

```
  <h5 class="text-muted mb-2">Checkbox Toggle Buttons (Independent States)</h5>
```

```
  <p class="small text-muted">Clicking these changes their visual focus state dynamically
without Javascript dependency.</p>
```

```
  <div class="btn-group" role="group" aria-label="Basic checkbox toggle button group">
    <input type="checkbox" class="btn-check" id="btncheck1" autocomplete="off">
    <label class="btn btn-outline-success" data-toggle-type="checkbox"
for="btncheck1">Toggle Option A</label>
```

```
    <input type="checkbox" class="btn-check" id="btncheck2" autocomplete="off">
    <label class="btn btn-outline-success" data-toggle-type="checkbox"
for="btncheck2">Toggle Option B</label>
```

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```
        <input type="checkbox" class="btn-check" id="btncheck3" autocomplete="off">
        <label class="btn btn-outline-success" data-toggle-type="checkbox"
for="btncheck3">Toggle Option C</label>
    </div>
</section>

</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Exercise 9: Buttons and Button Groups

Exploring styling variants, button clusters, and state control mechanisms.

Exercise 9.1: Contextual & Outline Buttons

Solid Theme Variants

Primary Secondary Success Danger Warning Info Light Dark Link

Outline Variants (.btn-outline-*)

Primary Secondary Success Danger Warning Info Dark

Exercise 9.2: Button Groups & Checkbox Toggles

Standard Button Group (.btn-group)

Left Middle Right

Checkbox Toggle Buttons (Independent States)

Clicking these changes their visual focus state dynamically without Javascript dependency.

Toggle Option A Toggle Option B Toggle Option C

11. Cards and Media Objects

Exercise 11.1:

Create a profile card using card, card-body, card-title, and card-img-top.

Exercise 11.2:

Design a media object layout using media and align an image to the left of the content.

Source Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Cards and Media Objects</title>

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="bg-light">

  <div class="container my-5">
```

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```
<header class="text-center mb-5">
  <h1 class="display-5 fw-bold">Exercise 11: Cards & Media Objects</h1>
  <p class="text-muted">Building rich content containers and component-aligned layouts.</p>
  <hr>
</header>

<div class="row g-5">

  <div class="col-md-5">
    <h2 class="h4 mb-3 text-secondary">Exercise 11.1: Profile Card</h2>

    <div class="card shadow-sm" style="max-width: 18rem;">

      <div class="card-body">
        <h5 class="card-title">Alex Morgan</h5>
        <h6 class="card-subtitle mb-2 text-muted">AI Engineer</h6>
        <p class="card-text">Passionate about training deep learning models, optimizing
neural networks, and developing computer vision solutions.</p>
        <a href="#" class="btn btn-primary btn-sm">View Portfolio</a>
        <a href="#" class="btn btn-outline-secondary btn-sm">Contact</a>
      </div>
    </div>

  <div class="col-md-7">
    <h2 class="h4 mb-3 text-secondary">Exercise 11.2: Media Object Layout</h2>
    <p class="small text-muted">Left-aligned image with text flowing smoothly alongside it
(Bootstrap 5 Flexbox implementation).</p>

    <div class="p-4 bg-white border rounded shadow-sm">

      <div class="d-flex align-items-start">

        <div class="flex-shrink-0">
          
        </div>

        <div class="flex-grow-1">
          <h5 class="mt-0 fw-bold">Nested System Notification</h5>
          <p class="mb-1 text-muted text-secondary">The latest package updates have been
successfully pushed to the main branch workspace. All unit diagnostics passed initialization tests
without reporting exceptions.</p>
          <small class="text-black-50">Received 3 minutes ago via GitHub
Webhook</small>
        </div>
      </div>
    </div>
  </div>
</div>
```

BOOTSTRAP

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Exercise 11: Cards & Media Objects

Building rich content containers and component-aligned layouts.

Exercise 11.1: Profile Card

User Profile Image

KM Sai Sowmya
AI Engineer
Passionate about training deep learning models, optimizing neural networks, and developing computer vision solutions.

[View Portfolio](#) [Contact](#)

Exercise 11.2: Media Object Layout

Left-aligned image with text flowing smoothly alongside it (Bootstrap 5 Flexbox implementation).

Media Asset Thumbnail

Nested System Notification

The latest package updates have been successfully pushed to the main branch workspace. All unit diagnostics passed initialization tests without reporting exceptions.

Received 3 minutes ago via GitHub Webhook